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When Education Expenditure Matters: An Empirical Analysis of Recent International Data

EMILIANA VEGAS AND CHELSEA COFFIN

We analyze the diminishing correlations between education expenditure and learning outcomes to address two fundamental questions: Do education systems with different levels of education spending have different student achievement levels? If so, at what amount of education spending does the relationship between increased expenditure and student achievement differ? Using data from a large group of countries around the world, we find that the association between education spending and student performance in mathematics is statistically significant among systems that spend below a threshold of US$8,000 per student annually (in purchasing power parity). Controlling for average income (GDP) per capita and income inequality, our estimates suggest that education spending is associated with increased student performance only among systems that spend below this threshold, with mean student achievement approximately 14 points higher on the PISA scale for every additional US$1,000 spent.

Introduction

Education systems should provide adequate resources to ensure that all students have the opportunity to receive a high-quality basic education (Berne and Stiefel 1984; Underwood 1995). Although standards of student achievement, as well as the costs to reach those standards, may vary across countries, we propose that there may be a minimum amount of resources required to bring about learning outcomes and provide essential inputs such as teachers, school buildings, and learning materials. If this is the case, we suggest that gains in learning outcomes may be associated with increases in education spending for countries that spend below this minimum amount, whereas gains in learning outcomes for those countries that spend above this cutoff may be associated with increased efficiency of spending instead. The precise minimum amount of resources has proved difficult to estimate, in great part because there is no direct, causal relationship between increased spending and improved learning outcomes.

In fact, counter to reasoning that resources are necessary for learning, a review of education finance literature for high-spending systems establishes that, in many instances, money doesn't matter for learning outcomes.

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Hanushek (1986)'s well-known analysis of relevant studies tabulated the significance and direction of estimated coefficients from 147 education production functions in published studies to establish which inputs matter for learning outcomes. Citing a preponderance of insignificant effects or inconsistent direction of effects, Hanushek found that there is no "strong or systematic relationship between school expenditures and student performance" (Hanushek 1986, 1162). This statement sparked rebuttals from Hedges et al. (1994) and Kremer (1995), who critiqued Hanushek's vote-counting methodology and endorsed a closer association between resources and outcomes. However, none of these analysts recommends an unconditional increase in spending. Hanushek (1994) admitted that money matters sometimes, Hedges et al. conceded that "throwing money" at schools is not best practice, and Kremer advised that increasing spending "typically improves student performance modestly" (1995, 247).

Studies that do examine the relationship between funding and results in developing countries that are often low spenders suggest that certain inputs (and by extension, some education expenditure) have the potential for a higher impact in these settings. Fuller and Clarke (1994) examined specific education inputs in their summary of over 100 education production function studies from developing countries, finding consistent effects for textbooks, teacher qualities, and instructional time, among others. An updated survey of this literature from Glewwe et al. (2011) finds that in developing countries, a few basic inputs do have significant effects, including availability of desks, teacher knowledge of the subjects they teach, and teacher absence. Adding to the discussion, a recent review of what works in education in developing countries by Krishnaratne et al. (2013) finds positive effects on test scores for better infrastructure building, educational materials, and additional teaching resources.

Even within the United States (a high spender), some studies suggest that money matters for states that spend below the national average. Ferguson and Ladd (1996) studied student outcomes at the student and district level in Alabama and found significant effects on learning outcomes not only for particular inputs (teacher test scores, teacher education, and class size) but also for instructional spending levels more generally. Consistent with the message of this study, they explained that these large predicted spending effects would not be expected in other states that were historically higher spenders, and that spending alone is not sufficient to increase student learning. Overall, the literature indicates that expenditure can be effective in most settings if managed efficiently, and more importantly for this study, may have a greater impact in developing countries or others that are likely to be low spenders.

Some studies look beyond the relationship of education expenditure and learning outcomes to investigate the importance of family background

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relative to school factors. Heyneman and Loxley's (1983) review of school and teacher qualities survey information found that school and teacher quality are more influential in predicting science learning outcomes than family background in low-income countries, which tend to be low spenders. However, Riddell (1989) critiqued the Heyneman-Loxley effect's methodology and data. Then, using a new technique of hierarchical linear modeling (HLM) and updated data from the 1990s, Baker et al. (2002) found that the level of national income is not negatively related to larger school effects. Hanushek and Luque's (2003) analysis of data from the 1995 TIMSS assessment confirmed the disappearance of the Heyneman-Loxley effect, finding that learning outcomes are not more positively related to school resource differences in poor countries. However, Chudgar and Luschei's (2009) HLM analysis of TIMSS 2003 results for grade 4 (a point in time when a more representative population of children remain in school in developing countries) shows that importance of schools is negatively related to a country's national income, similar to the Heyneman-Loxley effect. This study will further examine the importance of school resources in countries that spend less on education and tend to be poor.

In this study, we approach the correlational relationship between expenditure and outcomes by decomposing the education production function for high- and low-spending education systems. Our findings suggest that increased spending is associated with improved learning outcomes in low-spending systems up to a cutoff point of expenditure, beyond which there is no clear, statistically significant association between spending and outcomes in high-spending systems.

A scatter plot of spending and outcomes in mathematics[1] for our sample[2] suggests that the relationship between spending and outcomes does in fact differ between countries that spend more and less per student (see fig. 1). Indeed, the association is strong and statistically significant among countries with low education spending levels. The reason for this difference in the association between student achievement levels and per pupil expenditures at low and high levels of spending could be diminishing returns: as education spending increases, benefits at the margin tend to become low, suggesting a log-expenditures specification. Interestingly, income levels do not necessarily predict performance, noted by a recent Program for International Student Assessment (PISA) in Focus brief (OECD 2012) that found per capita GDP to be unrelated with gains in reading scores for countries with per capita GDP above US$20,000. Like this analysis, the PISA

[1] Mathematics has been chosen as the primary subject of comparison.

[2] The sample includes all countries with available data for PISA assessments and education expenditure per secondary school pupil (for the year 2010 or closest year available).

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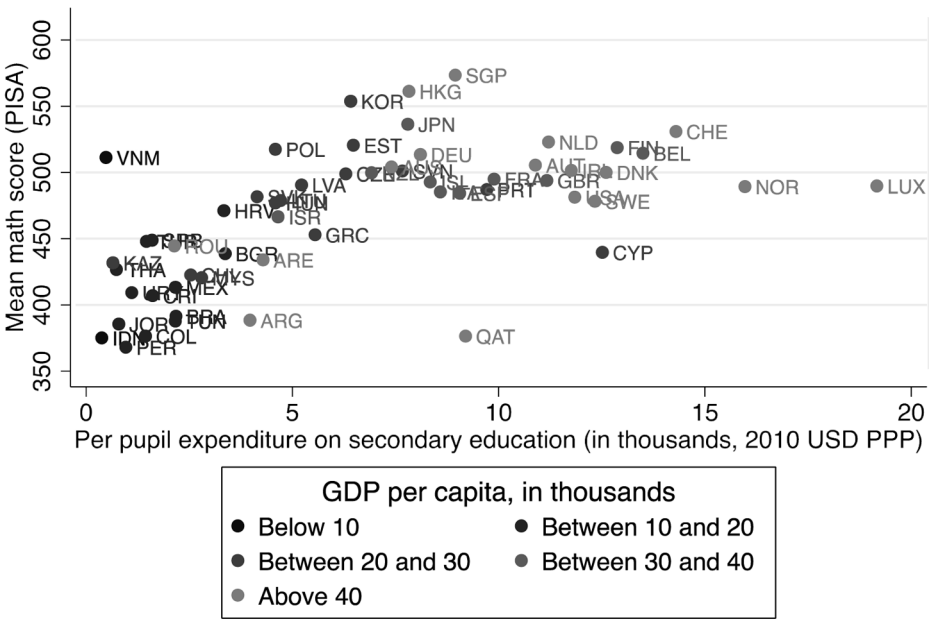

FIG. 1.—Mean score in math and annual education spending. SOURCE.—Authors' analysis.

brief illustrated that cumulative education expenditures[3] are not associated with reading scores in PISA after US$35,000 per student. In figure 1, large variations in test scores within high-income countries reinforce this finding. In addition, these figures show overlap in test scores between lower middle income and upper middle incomes countries, suggesting that income levels are not directly related to learning outcomes for these countries either.

A visual assessment and regression results in table 1 for mathematics suggest a logarithmic relationship between expenditure and outcomes over the whole sample. However, based on a review of the literature, we choose instead to investigate two groups of spenders so that we can identify the point at which increased spending will no longer be associated with gains in test scores. Separating the sample into high and low spenders will also allow us to better describe the two distinct relationships that occur in practice, starting with a steep slope among low spenders such as Indonesia or Jordan, and then flattening out for high spenders such as Germany or Japan. The threshold of education expenditure per pupil where the association be-

[3] Cumulative education expenditure is calculated as the amount a country spends on education for each student from age 6 to 15, converted into the equivalent international currency.

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TABLE 1
MEAN SCORES IN MATHEMATICS 2009 AND LOG EDUCATION EXPENDITURE

	Mathematics, without Controls (1)	Mathematics, with Controls (2)
Log of education expenditure	42.11***	44.04***
	(4.60)	(8.47)
Log of per capita GDP		−31.59
		(21.28)
Gini coefficient		−2.60***
		(.39)
Constant	111.95***	502.63***
	(38.80)	(159.16)
Observations	55	50
R-squared	.576	.792

NOTE.—Robust standard errors in parentheses.
SOURCE.—Authors' calculations.
*** $p < .01$.

tween student achievement levels and per pupil expenditures becomes statistically insignificant appears to be around US$8,000, but careful analysis will provide evidence to establish this location.

There is no accepted level of spending that determines a cutoff for when education expenditure matters. However, few would argue that spending on education does not contribute to learning, or that expenditure on education should be eliminated. Therefore, if zero spending is unreasonable and spending too much does not systematically improve outcomes, raising efficient education spending must matter until a certain threshold is reached at which adequate resources are provided and beyond which additional resources cease to matter.

To be clear, this study does not advocate an unconditional increase in spending. We agree that *how* money is spent is more important than *how much* money is spent, to a degree. Pritchett and Filmer (1999) explain this dual focus: it is possible for a school to increase efficiency and produce higher learning outcomes with less funding, but if two schools have the same efficiency in allocation, the school with the higher budget will outperform the school with the lower budget, though with diminishing returns. Efficiency of use matters and varies for most educational inputs that are believed to influence student outcomes, such as class size. Class size is often at the center of education expenditure debates (and therefore a key input to evaluate) because the number of students per classroom is often directly related to resource allocation and because educators and parents often prefer smaller class size (Angrist and Lavy 1999). Studies in Bangladesh and India showed that smaller class sizes did not positively influence learning outcomes (Asadullah 2005; Banerjee et al. 2007), but recent efforts to decrease class size improved test scores in other low-income settings, including South Africa

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(Case and Deaton 1999) and Bolivia (Urquiola 2006). Class size and other additional resources do have the potential to influence learning outcomes to a certain extent if they change a child's school experience (Murnane and Willett 2011).

If, as research, observed data patterns, and common sense suggest, when used effectively increased expenditure is associated with higher learning outcomes among low-spending systems, can we determine where the cutoff lies in education expenditure associated with gains in learning? There are wide within- and cross-country variations in both learning outcomes and education spending per pupil, with low spenders typically achieving at lower levels. Even within a country, it is difficult to determine whether increased spending on education results in better outcomes because spending is not the only characteristic that affects test scores (Guryan 2001). The same argument is further complicated in cross-country comparisons with increased variation in background conditions. Education spending could be correlated with other factors that affect learning outcomes, creating omitted variable bias. This analysis will attempt to control for the influences of income and inequality, as consistent data across countries that addresses other factors such as child poverty rates, share of children from single parent homes, or malnutrition rates, are limited.

Data

The data used for our analysis come from international assessments of student learning and various international statistical databases. For a summary of descriptive statistics, see table 2 Learning outcomes are measured by mean score in mathematics and reading on PISA. PISA has been selected as the international assessment for this analysis as results for both mathematics and reading are available. The analysis examines results for all available years

TABLE 2
DESCRIPTIVE STATISTICS

	Full Sample
Mean mathematics scores	463.09
	(51.76)
Education expenditure per secondary pupil, in thousands	6.51
	(4.75)
Per capita GDP, in thousands	29.37
	(20.12)
Gini coefficient	34.45
	(9.21)
Observations	50

NOTE.—Standard deviations in parentheses. Mean scores refer to student outcomes on PISA international mathematics assessment for 2009. Expenditure and per capita GDP are expressed in thousands of international dollars.

SOURCE.—Authors' calculations.

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(2006, 2009, and 2012) in an effort to seek a consistent trend over time. Mathematics has been selected as the primary subject of comparison because it is recognized as one of the basic cognitive skills that are essential for full participation in society (OECD 2004).

The explanatory variable for learning outcomes is expenditure on secondary education per secondary school pupil reported in United Nations Educational, Scientific, and Cultural Organization (UNESCO) Institute for Statistics' Global Education Digest. Although it would be ideal to use per pupil cumulative education spending to the appropriate grade level, these data were only available for 30 systems of interest. Instead, education expenditure on secondary education per pupil (expressed in 2010 US dollars and adjusted for purchasing power parity) has been used for 51 systems, with data for the year 2010 or closest year available to the year of PISA administration. Expenditure on secondary education per pupil is a reasonable proxy for cumulative education expenditure: a straightforward correlation between the two is 0.92, and a regression of cumulative expenditure on secondary education expenditure yields an R^2 of 0.85 with a highly significant and large coefficient on secondary expenditure.

Control variables include the country's average level of income and degree of income inequality. Although these variables do not account for all unobserved characteristics that may nonetheless affect learning outcomes, it is hoped that some major factors (such as malnutrition or family size) will be represented by these controls. The level of income is indicated by GDP per capita (expressed in 2010 US dollars and adjusted for purchasing power parity) for the year 2010 or the closest year available to the date of PISA administration. The Gini coefficient indicates the national level of income inequality, and comes from different data sources depending on country status as developing (World Development Indicators, World Bank), OECD (OECD. Stat Extracts, Organization for Economic Cooperation and Development), or other (World Factbook, Central Intelligence Agency of the United States of America).

There are several issues with these data, including bias toward systems that participate in international assessments and report education expenditure to UNESCO, the possibility that education expenditure may need to be adjusted using a sector-specific price index (not simply purchasing power parity), and the lack of data on some unobserved characteristics that may affect learning outcomes. More generally, a small sample size will make the detection of statistically significant associations less likely than would be the case for a larger sample. Findings should be interpreted accordingly. In spite of these complications, taking advantage of the best available data is better than no analysis, and we hope that this study inspires more research concerning the impact of education expenditure on learning outcomes, particularly in developing countries.

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Methodology

In the real world—a factor that complicates our analysis—education systems have not been assigned to groups of low and high spenders by any established classification system, so we must look to the data themselves to identify the cutoff point in the range of expenditures, if in fact one exists. Because we cannot take advantage of a designed or natural experiment, our findings should be interpreted cautiously as associational, not causal. To investigate whether the relationship between spending and learning outcomes differs across systems that spend differently in education, we first differentiate high- from low-spending systems with indicator variables:

$$\text{Constrained: } S = \gamma_0 + \gamma_1 E_{\mathrm{i}} + \gamma_2 \mathrm{I}_{\mathrm{i}} + \gamma_3 G_{\mathrm{i}} + \eta \tag{1}$$

$$\text{Full: } S = \beta_0 + \beta_1 E_{\mathrm{i}} + \beta_2 I_{\mathrm{i}} + \beta_3 G_{\mathrm{i}} + \beta_4 H_{\mathrm{i}} + \beta_5 H_{\mathrm{i}} * E_{\mathrm{i}} + \beta_6 H_{\mathrm{i}} * I_{\mathrm{i}} + \beta_7 H_{\mathrm{i}} * G_{\mathrm{i}} + \varepsilon. \tag{2}$$

In these equations, S is standardized mean score on an international mathematics assessment, E is expenditure on secondary education per pupil, I is income level represented by GDP per capita, G is inequality measured by the Gini coefficient, H is an indicator variable to classify systems as high spenders if their education expenditure is greater than the designated cutoff explained below, and η and ϵ are error terms. A system is designated as a high spender if $E \geq H$, and as a low spender if $E < H$; γ_1, β_1, and β_5 are the coefficients of interest and show the association between per pupil expenditure and education outcomes.

To determine whether the estimated coefficients on expenditure differ for high- and low-spending systems, we use the Chow (1960) test, which detects structural change using an analysis of covariance for the constrained model (where all observations belong to one group) and full model (where observations belong to different groups). Typically, this test is used with panel data to see if a relationship holds stable over time, but we adapt it to assess whether the association between learning outcomes and per pupil expenditure remains the same as the amount of expenditure increases. We use and build on a dummy variable method (first developed by Dufour [1980]), which forms dummy variables for the second time period (high-spending systems in this case) and uses an F-test to evaluate whether their coefficients are equal to zero. If the coefficients are equal to or not statistically different from zero, then we can assume that there is no difference in outcomes for high and low spenders due to any of the dependent variables. We establish indicator variables and evaluate the following hypothesis:

$$\mathrm{H}_0\text{: } \beta_4, \beta_5, \beta_6, \text{ and } \beta_7 = 0 \mathrm{H}_a\text{:} \tag{3}$$

At least one coefficient is different from 0,

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where β_4 is the dummy variable for high-spending systems, β_5 is the interaction between education expenditure and high-spending systems, β_6 is the interaction between income level and high-spending systems, and β_7 is the interaction between inequality and high-spending systems.

We define a series of high- and low-spending groups by arbitrarily setting cutoffs in education expenditure at increments of US$1,000. Next, we verify where expenditure (β_5) is different for high and low spenders.

If there are different expected outcomes for high and low spenders, how can we identify the natural cutoff between spending groups? Instead of simply assuming where the cutoff lies from a visual assessment, we test for a cutoff by observing the level of spending at which there is a statistically different relationship for high and low spenders. For the pair of high and low spenders at each increment of US$1,000, we test to see if the coefficient β_5 in model (2) is statistically different from zero. More specifically, a system has the following estimated associations with education expenditure:

$$\begin{aligned} &\text{Low spenders: } \beta_1 \\ &\text{High spenders: } \beta_1 + \beta_5 \end{aligned} \tag{4}$$

The term β_1 indicates the association for low spenders, and β_5 indicates whether there is a different impact for high spenders. When β_5 (the coefficient on the interaction term for education expenditure and high spenders) is either positive or negative and statistically significant, two distinct relationships exist for high and low spenders. We observe at which cutoff β_5 is no longer statistically different from zero, because at that point there is no difference in the association between student achievement and education expenditure among high- and low-spending systems. If the coefficient on education expenditure for low-spending systems is significant at this point, this would imply that an increase in education expenditure is associated with an increase in learning outcomes at lower levels of expenditure, and that a similar increase is associated with no change in learning outcomes at higher levels of expenditure.

To find a consistent trend over time, this methodology was applied to six separate regressions with PISA mean scores in math and reading in the years 2006, 2009, and 2012 as the dependent variables. Finally, results across years and subjects were reviewed.

Results

From our analysis, we find that the relationship between education expenditure and learning outcomes does change at a certain cutoff point, indicating that additional education expenditure is associated with higher student learning levels among low spenders, up to a threshold amount. As

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previously described, we fit a series of linear regressions to reach this conclusion. For this discussion, results for the PISA math 2009 are highlighted, as they are the closest to the average findings across subjects and years.

First, we fit basic regressions of learning outcomes on education expenditure with and without the control variables, to see if the estimated coefficient on education expenditure is statistically significant in a model that includes all observations for high and low spenders with available data. A regression of model (1) with control variables and no differentiation between high and low spenders suggests that the relationship between education expenditure and mean test scores is statistically significant (see table 3). Results of a regression of model (1) without control variables imply that the estimated coefficient on education expenditure is statistically significant for learning outcomes in mathematics (see table 3). Next, we investigate the extent to which the relationship may differ between high- and low-spending systems.

Second, to assess whether there is a structural change in the relationship between student learning outcomes and per pupil spending between high- and low-spending systems, we perform Chow tests for a series of regressions with dummy cutoffs at increments of US$1,000. For reference, table 4 shows that the difference between the coefficient estimates on education expenditure, GDP per capita, and Gini coefficient for high-spending systems and those for low-spending systems are significantly different for all tested cutoffs. The residual sum of squares is lowest at US$8,000, suggesting a best fit in that range, where there is the lowest amount of variation from the line of best fit. However, we are more interested in the point when the coefficient estimate on education expenditure in particular begins to have a differential association with learning outcomes.

Third, we look more closely at the interaction of education expenditure and the dummy for high-spending systems to see when it is no longer statistically significant as expenditure increases. From figure 1, we expect to find a break somewhere after US$8,000. At this point, there will no longer be any difference between the coefficient estimates for high- and low-spending systems, and the association of education expenditure with student learning outcomes will not be different from zero among high spenders. In fitting a series of regressions with different cutoffs in education expenditure for high- and low-spending systems (see table 3), we find that the coefficient estimate on education expenditure is statistically significant in the full model (2) at some expenditure levels for both low- and high-spending systems. Fitting regressions at increments of US$1,000 shows that there is a different, positive, and statistically significant estimated association for high-spending systems on average across subjects and years of US$7,333. We present results for mathematics 2009 as the model closest to the finding across all models, with education expenditure significant up to US$8,000.

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TABLE 3

MEAN SCORES AND EDUCATION EXPENDITURE (MATHEMATICS, 2009)

Variables	Full Model (No Controls) (1)	Full Model (2)	1,000 (3)	2,000 (4)	3,000 (5)	4,000 (6)	5,000 (7)	6,000 (8)	7,000 (9)	8,000 (10)	9,000 (11)	10,000 (12)	11,000 (13)
Education expenditure	7.752***	8.733***	10.78	41.54***	8.563	10.11*	16.03***	15.90***	15.77***	14.56***	13.04***	13.60***	12.77***
	(1.2)	(1.113)	(38.30)	(10.86)	(13.68)	(5.751)	(4.137)	(3.267)	(2.503)	(2.583)	(2.772)	(2.069)	(1.946)
Per capita GDP		−1.062***	2.496	2.629	1.478	1.487	.723	.288	.306	.374	.176	−1.205***	−1.166***
		(.141)	(2.509)	(1.786)	(1.804)	(1.576)	(1.378)	(1.110)	(.985)	(.806)	(.788)	(.243)	(.226)
Gini coefficient		−2.318***	−1.217	−1.886***	−1.312*	−1.408***	−1.698***	−1.731***	−1.660***	−1.586***	−1.671***	−2.112***	−2.108***
		(.440)	(1.102)	(.483)	(.688)	(.386)	(.298)	(.293)	(.287)	(.291)	(.308)	(.359)	(.368)
High spender constant			156.7***	152.6***	105.9**	109.1***	90.93**	80.26*	94.08**	78.42	18.67	−37.12	−43.80
			(48.71)	(36.80)	(41.44)	(36.14)	(41.46)	(41.98)	(45.49)	(53.52)	(54.19)	(72.85)	(110.4)
Education expenditure * High spender			−4.519	−35.40***	−2.766	−5.085	−11.38**	−11.71***	−11.61***	−9.007***	−5.268	−4.604	−3.362
			(38.31)	(10.92)	(13.72)	(5.848)	(4.385)	(3.597)	(3.046)	(3.224)	(3.360)	(6.304)	(9.141)
Per capita GDP * High spender			−3.469	−3.629**	−2.634	−2.686*	−1.948	−1.563	−1.527	−1.602*	−1.495*	−.488	−.580
			(2.513)	(1.791)	(1.810)	(1.583)	(1.391)	(1.132)	(1.007)	(.823)	(.802)	(.916)	(1.261)
Gini * High spender			−1.763	−.981	−.321	−.0368	.572	.980	.454	.336	1.248	2.364*	2.429
			(1.172)	(.644)	(.953)	(.755)	(.971)	(1.110)	(1.150)	(1.270)	(1.231)	(1.374)	(1.775)
Constant	412.593***	517.3***	402.6***	405.6***	426.2***	428.3***	442.7***	450.5***	447.6***	446.1***	456.1***	495.2***	496.8***
	(9.03)	(20.54)	(44.58)	(31.56)	(35.96)	(28.92)	(25.30)	(22.68)	(20.90)	(18.00)	(18.57)	(17.08)	(17.48)
Observations	50	50	50	50	50	50	50	50	50	50	50	50	50
R-squared	.506	.738	.810	.812	.843	.867	.861	.861	.863	.872	.863	.814	.801

NOTE.—Robust standard errors in parentheses.

* $p < .1$.

** $p < .05$.

*** $p < .01$.

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TABLE 4
STRUCTURAL CHANGE TEST FOR MEAN SCORE MODEL, MATHEMATICS

Cutoff Point	Numerator Degrees of Freedom	Denominator Degrees of Freedom	*F*-test	*p*-value	Sum of Squared Residuals
1,000	4	42	9.0	.000	24,896.8
2,000	4	42	8.0	.000	24,624.6
3,000	4	42	7.2	.000	20,600.7
4,000	4	42	12.7	.000	17,414.9
5,000	4	42	11.0	.000	18,286.8
6,000	4	42	11.1	.000	18259.7
7,000	4	42	12.7	.000	18,001.9
8,000	4	42	15.8	.000	16,790.1
9,000	4	42	10.1	.000	17,994.5
10,000	4	42	6.4	.000	24,428.3
11,000	4	42	5.7	.001	26,058.3

SOURCE.—Authors' calculations.

Estimations for the mathematics model are the following:

$$\begin{aligned} S &= 446.1 + 14.56E + 0.374I - 1.586G + 78.42H - 9.007H \\ &\quad * E - 1.602H * I + 0.336H * G \\ &\quad (18)(2.583)(0.806)(0.291)(53.52)(3.224)(0.823)(1.27) \\ &\quad H = 1 \text{ if } E > 8 \end{aligned} \tag{5}$$

These are then the implications for high and low spenders:

$$\begin{aligned} \text{High: } S_{E>8} &= 524.5 + 5.552E - 1.228I - 1.251G \\ &\quad (51.99)(1.991)(0.170)(1.275) \\ \text{Low: } S_{E<8} &= 446.1 + 14.56E + 0.374I - 1.586G \\ &\quad (17.67)(2.536)(0.791)(0.285) \end{aligned} \tag{6}$$

In the mathematics model, the estimated association between student learning outcomes and education expenditure is positive for low-spending systems (those that spend up to US$8,000 per student), while it is negative for high-spending systems (those that spend US$8,000 or more per student). Controlling for average income and its inequality, every additional US$1,000 in per pupil spending is associated with an improvement in mean scores of 14.56 points for low-spending systems up to the cutoff of US$8,000. Among low-spending and high-spending systems, the association between the control variable of income inequality and student learning outcomes is statistically significant and negative, indicating that unequal distribution of income is linked to performance.

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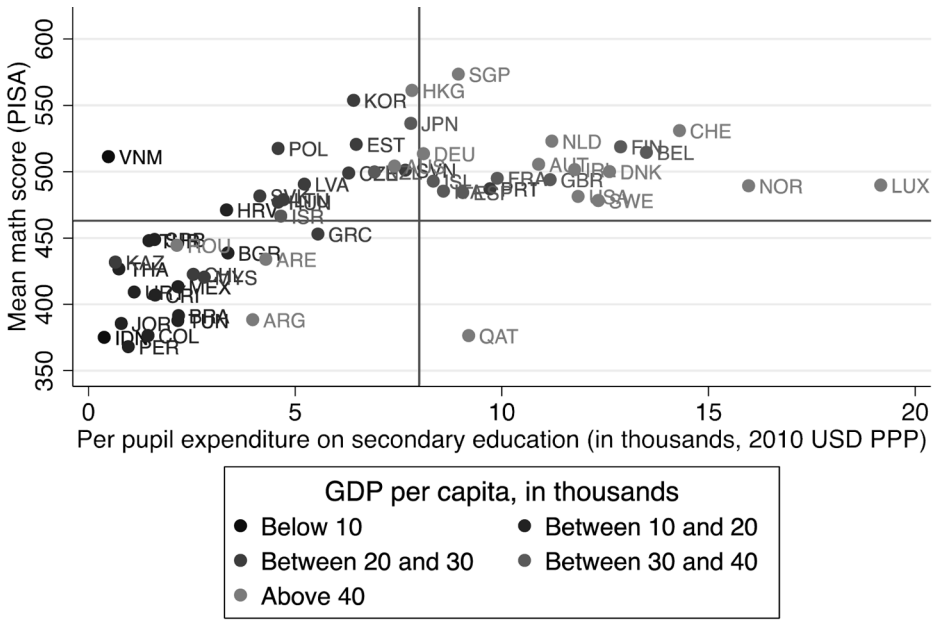

FIG. 2.—Mean score in math and annual education spending. SOURCE.—Authors' analysis.

Clearly, issues of efficiency and adequacy are related. Education systems can be roughly categorized in a 2 × 2 matrix of (1) high spender, high learning outcomes; (2) high spender, low learning outcomes; (3) low spender, low learning outcomes; and (4) low spender, high learning outcomes (see fig. 2).

Using cutoffs of the average of 463 in mean score, we observe only one high-spending system that is a low performer in mean score. We do observe a cluster of low-spending systems that are low performers as well as another cluster of systems that are high performers. This approximate positioning supports our findings that a certain amount of spending does appear to significantly impact adequate learning outcomes. Future analysis could involve a closer look at the characteristics of education systems for countries that score above the mean but spend relatively less.

Conclusion

Our estimates of the association between education expenditures and mean student learning outcomes in mathematics suggest that, when education systems spend above US$8,000, the association is no longer statistically significant. Above this level of per student expenditure, additional

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spending does not have a significant estimated association with learning outcomes. Therefore, we find a threshold effect after this level of resources is met, indicating a declining relationship between resources and achievement at high levels of expenditure (consistent with other recent literature).

One interpretation of our analysis, consistent with prior studies, is that efficiency in use of the resources spent on education is more likely to be associated with higher levels of learning outcomes among systems that already provide the basic inputs necessary for a quality education (as measured by their average spending per pupil). High spenders might also spend more on programs that compensate for students with disadvantaged backgrounds, which could produce inconsistent gains in test scores or proficiency, depending on the student. However, at expenditure levels below the estimated thresholds, there is a positive, statistically significant association between education expenditure and test score levels. When low spenders increase expenditure, it may be used to establish basic conditions or increase quality to a minimum standard, although efficient use of these resources may also be a constraint to achieve high levels of learning for all.

Despite our careful analysis, we should stress that the demonstrated association between education expenditure and spending cannot be assumed to be causal. Although we have controlled for some observables that affect learning outcomes, the analysis is restricted by availability of data, which makes it impossible to account for all influences on education. As more data become available, the analysis could be developed further with a fixed effects model, using the difference in spending (ideally cumulative education expenditure to the age of international assessment) and the difference in learning outcomes over time as explanatory and dependent variables, respectively. In this case, other country-specific factors would be controlled for, and a causal relationship could be established. Other future research could examine variations at the local level in spending and learning outcomes, adjusting for student needs, for a single low-spending country.[4] This would also reduce issues related to cross-country comparisons. More feasibly, the methodology used in the study could evaluate learning outcomes in regional assessments such as the Latin American Laboratory for Assessment of the Quality of Education (LLECE), the Southern and Eastern Africa Consortium for Monitoring Education Quality (SACMEQ), or Program on the Analysis of Education Systems (PASEC). As more countries participate in the PISA assessment and report their expenditure data to UNESCO, the existing data set will improve as well. We hope that future research will deepen our understanding of how money may matter for learning outcomes in systems that spend little.

[4] See Boser (2011) for an example of this type of evaluation in a high-spending country (the United States).

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